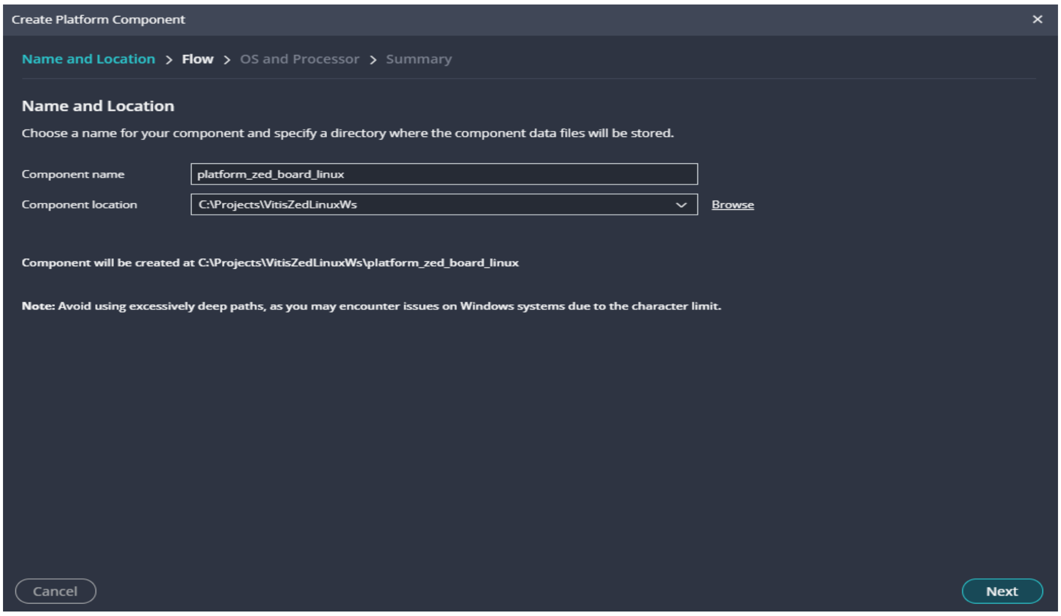
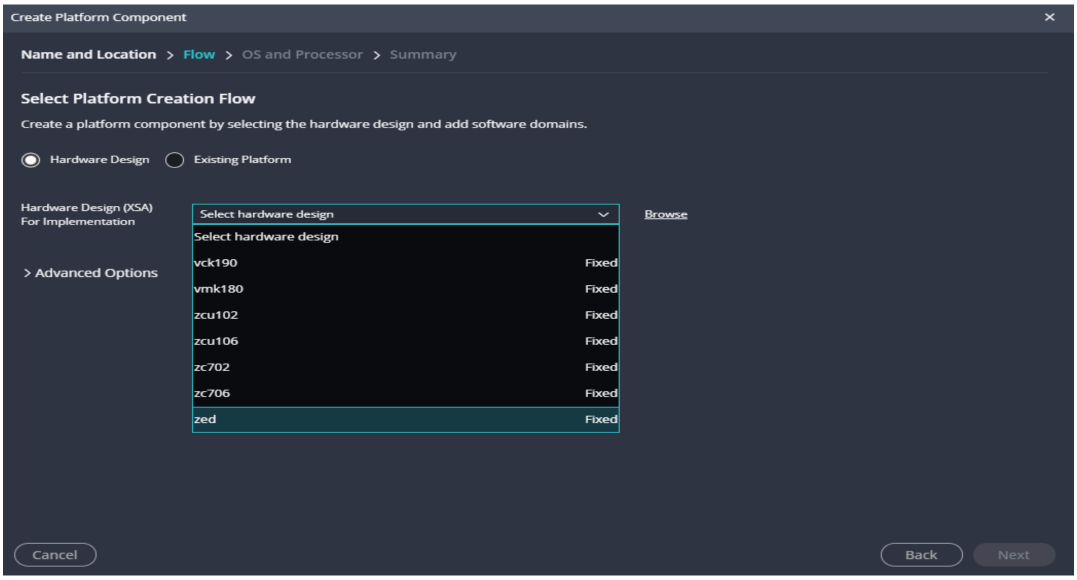
Build and debug 'Linux Hello World' with Vitis Unified

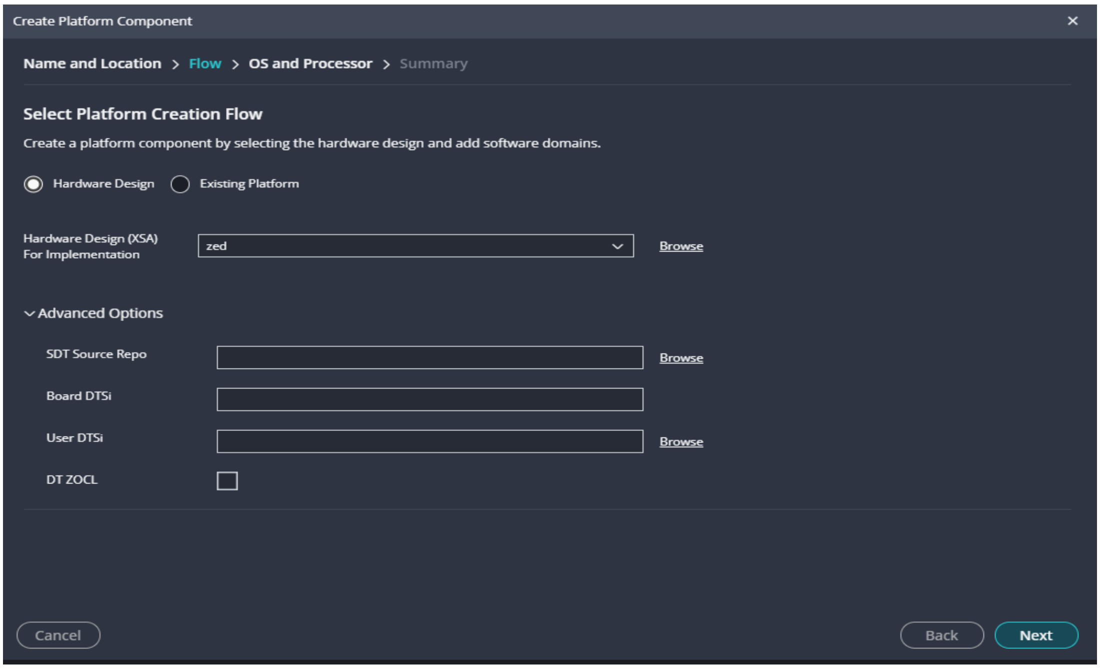
- Want to run/debug a linux application on our ZedBoard
- Will use the 'linux hello world' example project that comes with Vitis

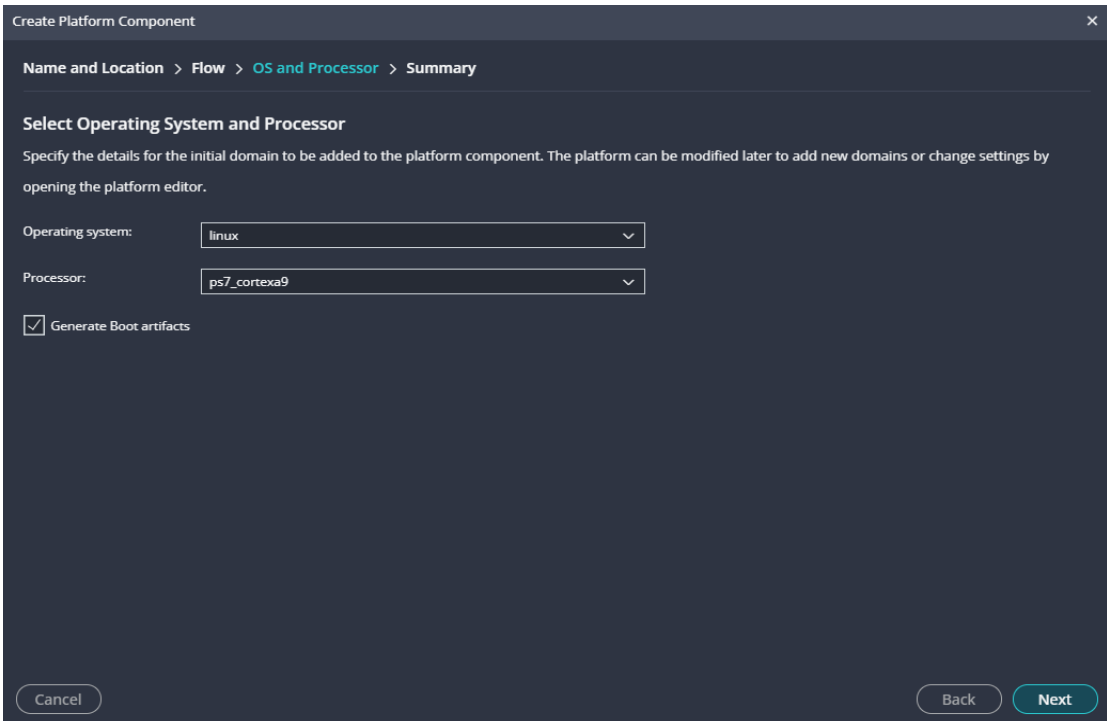
Using Vitis Unified 2024.2:

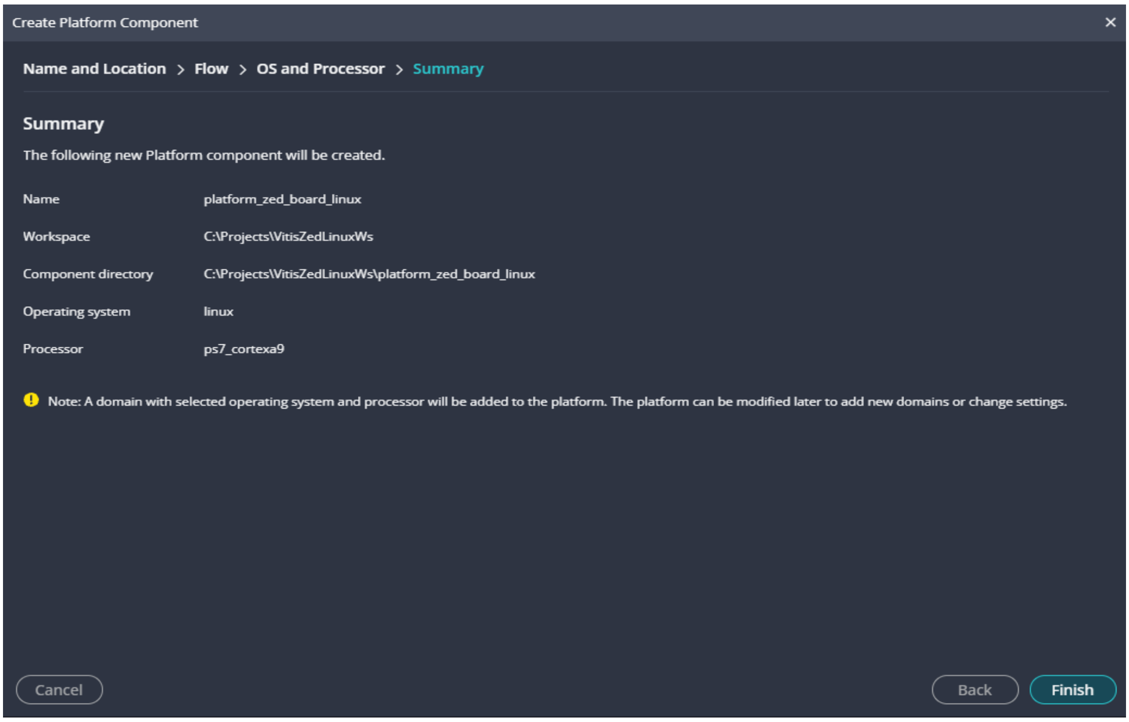
1. Set workspace to 'Projects/VitisZedLinuxWs' (brand new folder)
2. In 'Embedded Development' panel, click 'Create Platform Component'






Create Platform Component
Name and Location > Flow > OS and Processor > Summary
Select Platform Creation Flow
Create a platform component by selecting the hardware design and add software domains.
Hardware Design
Existing Platform
Hardware Design (XSA) For Implementation
zed
Browse
Advanced Options
SDT Source Repo
Browse
Board DTSi
User DTSi
Browse
DT ZOCL
Cancel
Back
Next


Create Platform Component
Name and Location > Flow > OS and Processor > Summary
Select Operating System and Processor
Specify the details for the initial domain to be added to the platform component. The platform can be modified later to add new domains or change settings by opening the platform editor.
Operating system:
linux
Processor:
ps7_cortexa9
Generate Boot artifacts
Cancel
Back
Next

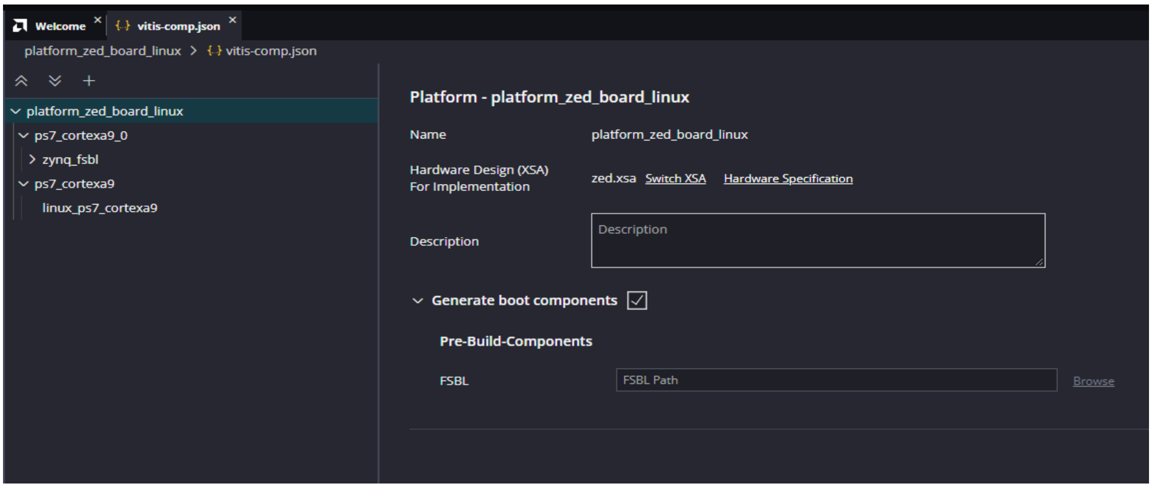


Output console:

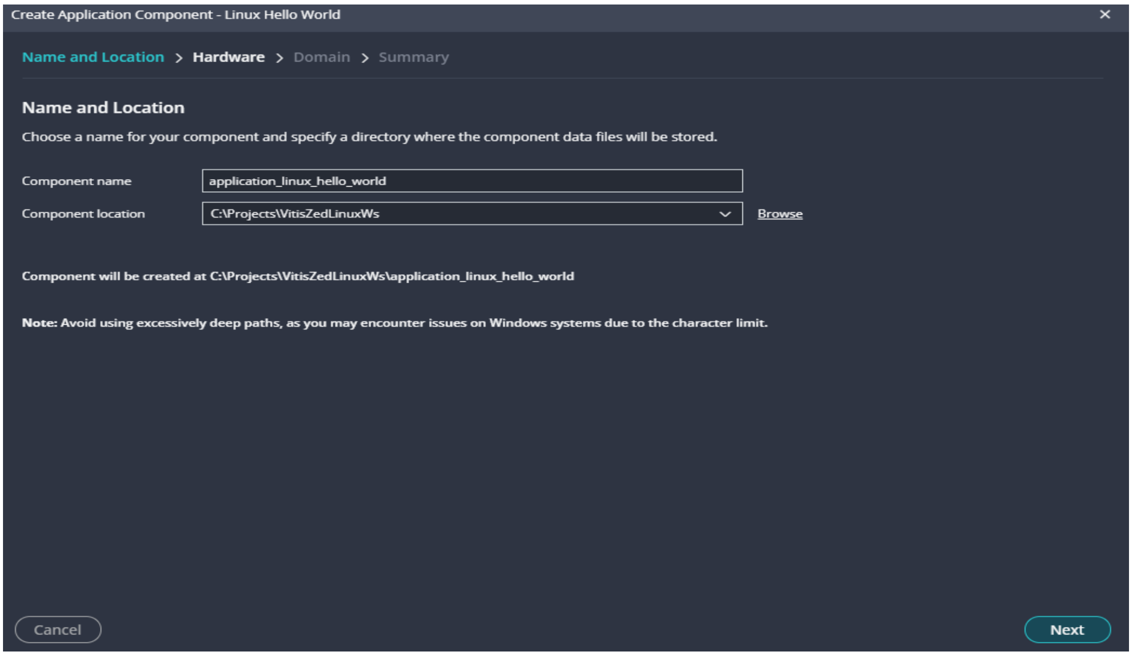
```
18:01:33 INFO  : updated workspace path: C:\Projects\VitisZedLinuxWs
18:01:35 INFO  : Successfully created repository data at 
C:\Users\wlaba\AppData\Local\Temp\rigel_lopper_wlaba15563819171094951191
18:01:35 INFO  : cmd.exe, /C, C:\Xilinx\Vitis\2024.2\tps\win64\lopper-
1.1.0\env\Scripts\activate.bat && python 
C:\Xilinx\Vitis\2024.2\data\embeddedsw\scripts\pyesw\\get_template_data.py -d 
C:\Users\wlaba\AppData\Local\Temp\rigel_lopper_wlaba11976232445309139885 -r 
C:\Users\wlaba\AppData\Local\Temp\rigel_lopper_wlaba15563819171094951191\.repo.yaml 
&& C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\deactivate.bat
18:01:36 INFO  : Embedded Template List generated successfully
18:01:36 INFO  : cmd.exe, /C, C:\Xilinx\Vitis\2024.2\tps\win64\lopper-
1.1.0\env\Scripts\activate.bat && python 
C:\Xilinx\Vitis\2024.2\data\embeddedsw\scripts\pyesw\\repo.py -st  
C:\Xilinx\Vitis\2024.2\data\embeddedsw && C:\Xilinx\Vitis\2024.2\tps\win64\lopper-
1.1.0\env\Scripts\deactivate.bat
18:01:38 INFO  : Successfully created repository data at 
C:\Projects\VitisZedLinuxWs/_ide\.wsdata
18:01:38 INFO  : cmd.exe, /C, C:\Xilinx\Vitis\2024.2\tps\win64\lopper-
1.1.0\env\Scripts\activate.bat && python 
C:\Xilinx\Vitis\2024.2\data\embeddedsw\scripts\pyesw\\get_template_data.py -d 
C:\Users\wlaba\AppData\Local\Temp\rigel_lopper_wlaba8550755919004480838 -r 
C:\Projects\VitisZedLinuxWs\_ide\.wsdata\.repo.yaml && 
C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\deactivate.bat
18:01:39 INFO  : Embedded Template List generated successfully
18:40:59 INFO  : Found no platform with name 'zed' in install repositories

18:40:59 INFO  : Install SDT for given XSA copied Successfully
```

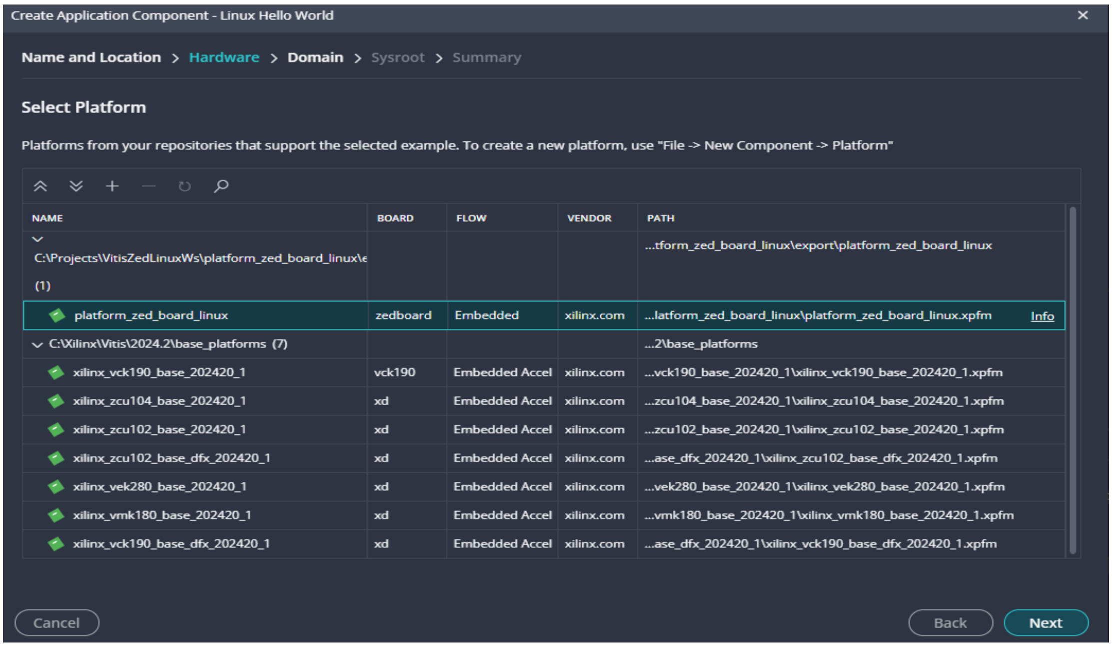
```
18:40:59 INFO  : cmd.exe, /C, C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\activate.bat && lopper --enhanced --werror -f -O C:\Projects\VitisZedLinuxWs\.rigel_lopper -i C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Lib\site-packages\lopper\lops\lop-cpu-oslist.dts C:\Projects\VitisZedLinuxWs\.rigel_lopper_temp_platform\zed\hw\sdt\system-top.dts && C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\deactivate.bat

18:41:01 INFO  : CPU List generated successfully

18:47:13 INFO  : Platform platform_zed_board_linux creation started.

18:47:13 INFO  : Lopper command for cpu List generation: lopper --enhanced --werror -f -O C:\Projects\VitisZedLinuxWs\.rigel_lopper -i C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Lib\site-packages\lopper\lops\lop-cpu-oslist.dts C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\hw\sdt\system-top.dts

18:47:13 INFO  : cmd.exe, /C, C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\activate.bat && lopper --enhanced --werror -f -O C:\Projects\VitisZedLinuxWs\.rigel_lopper -i C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Lib\site-packages\lopper\lops\lop-cpu-oslist.dts C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\hw\sdt\system-top.dts && C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\deactivate.bat

18:47:14 INFO  : CPU List generated successfully

18:47:14 INFO  : created .gitignore file for the project C:\Projects\VitisZedLinuxWs\platform_zed_board_linux

18:47:14 INFO  : ZYNQ: Using the QEMU args from install at  : C:\Xilinx\Vitis\2024.2\data\emulation\platforms\zynq\sw\a9_linux\qemu\qemu_args.txt

18:47:14 INFO  : ZYNQ: Using the QEMU args from install at  : C:\Xilinx\Vitis\2024.2\data\emulation\platforms\zynq\sw\a9_standalone\qemu\qemu_args.txt

18:47:14 INFO  : ZYNQ: Using the QEMU args from install at  : C:\Xilinx\Vitis\2024.2\data\emulation\platforms\zynq\sw\a9_linux\qemu\qemu_args.txt

18:47:14 INFO  : Domain linux_ps7_cortexa9 added successfully.

18:47:14 INFO  : lopper command to generate BSP :python C:\Xilinx\Vitis\2024.2\data\embeddedsw/scripts/pyesw//create_bsp.py -w C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\zynq_fsbl\zynq_fsbl_bsp -p ps7_cortexa9_0 -o standalone -s C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\hw\sdt\system-top.dts -t zynq_fsbl -r C:\Projects\VitisZedLinuxWs\_ide\.wsdata\.repo.yaml

18:47:14 INFO  : cmd.exe, /C, C:\Xilinx\Vitis\2024.2\tps\win64\lopper-
```

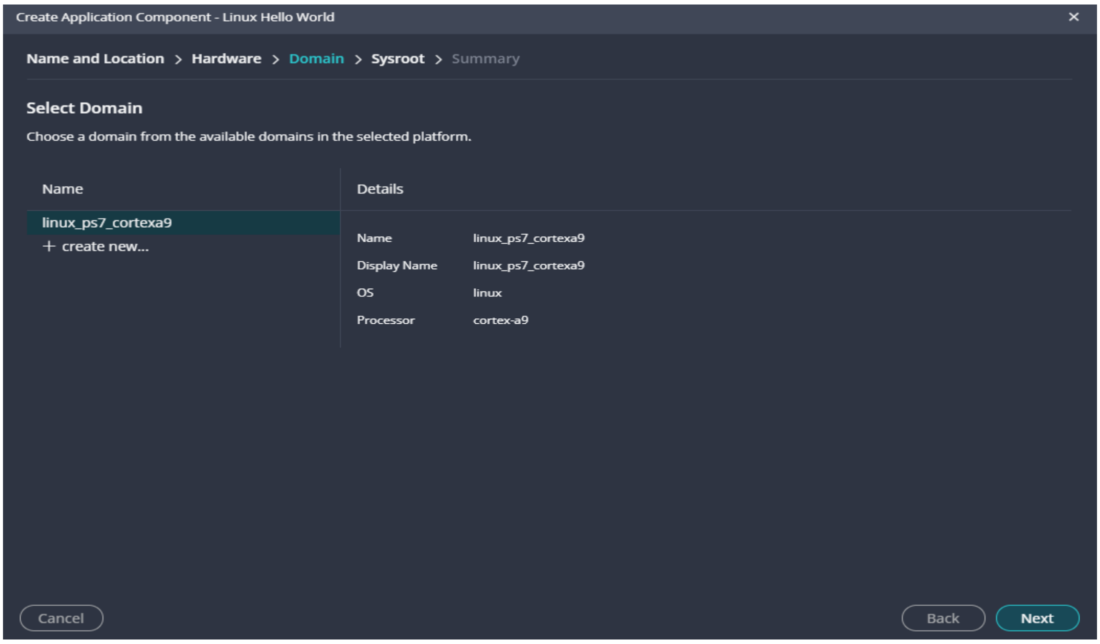
```
1.1.0\env\Scripts\activate.bat && python
C:\Xilinx\Vitis\2024.2\data\embeddedsw\scripts\pyesw\\create_bsp.py -w
C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\zynq_fsbl\zynq_fsbl_bsp -p
ps7_cortexa9_0 -o standalone -s
C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\hw\sdt\system-top.dts -t
zynq_fsbl -r C:\Projects\VitisZedLinuxWs\_ide\.wsdata\.repo.yaml &&
C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\deactivate.bat

18:47:29 INFO  : Successfully created Domain at
C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\zynq_fsbl\zynq_fsbl_bsp

18:47:29 INFO  : lopper command to create baremetal application :python
C:\Xilinx\Vitis\2024.2\data\embeddedsw/scripts/pyesw//create_app.py -s
C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\zynq_fsbl -t zynq_fsbl -d
C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\zynq_fsbl\zynq_fsbl_bsp -n
fsbl -r C:\Projects\VitisZedLinuxWs\_ide\.wsdata\.repo.yaml

18:47:29 INFO  : cmd.exe, /C, C:\Xilinx\Vitis\2024.2\tps\win64\lopper-
1.1.0\env\Scripts\activate.bat && python
C:\Xilinx\Vitis\2024.2\data\embeddedsw\scripts\pyesw\\validate_bsp.py -t zynq_fsbl
-d C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\zynq_fsbl\zynq_fsbl_bsp -r
C:\Projects\VitisZedLinuxWs\_ide\.wsdata\.repo.yaml &&
C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\deactivate.bat

18:47:32 INFO  : Successfully validated template zynq_fsbl

18:47:37 INFO  : Successfully Created Application sources at
C:/Projects/VitisZedLinuxWs/platform_zed_board_linux/zynq_fsbl

18:47:37 INFO  : Platform FSBL Boot domain added successfully.

18:47:38 INFO  : Platform creation finished successfully.

18:47:38 INFO  : Generating Export directory

18:47:38 INFO  : Generated platform metadata for creating application(s) based on
platform platform_zed_board_linux.
```

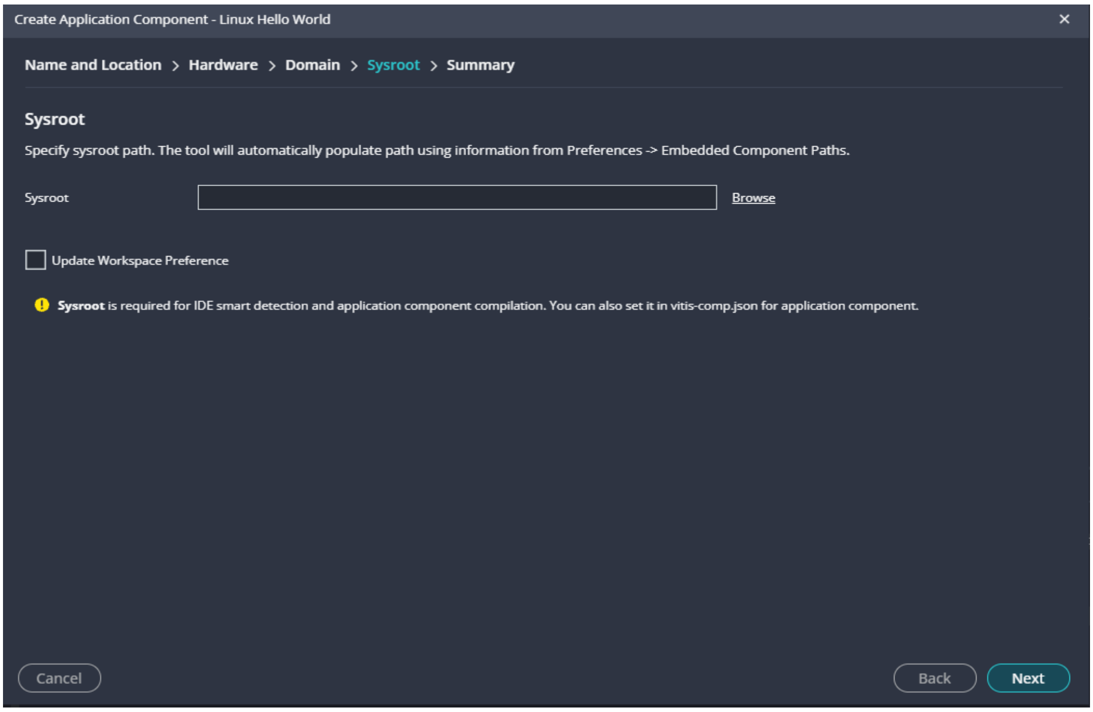


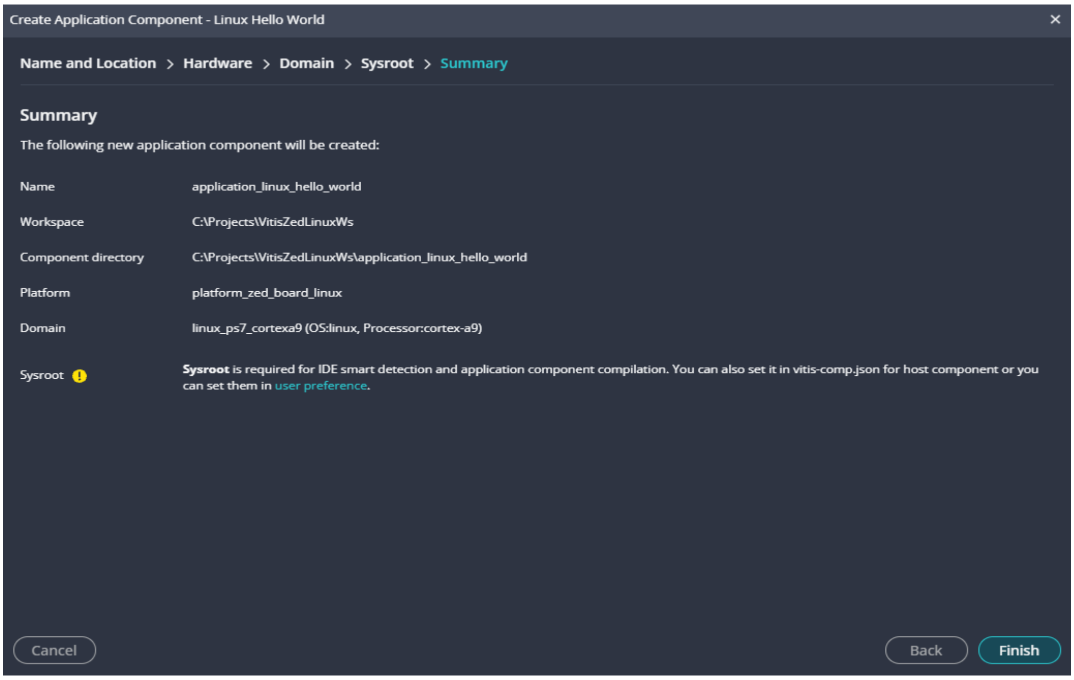
3. Click on Examples Icon and select '+' sign to right of 'Linux Hello World' to create 'New Application Component'




Create Application Component - Linux Hello World
Name and Location > Hardware > Domain > Sysroot > Summary
Select Platform
Platforms from your repositories that support the selected example. To create a new platform, use "File -> New Component -> Platform"
NAME
BOARD
FLOW
VENDOR
PATH
C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\e
(1)
...tform_zed_board_linux\export\platform_zed_board_linux
platform_zed_board_linux zedboard Embedded xilinx.com ...latform_zed_board_linux\platform_zed_board_linux.xpfm Info
C:\Xilinx\Vitis\2024.2\base_platforms (7) ...2\base_platforms
xilinx_vck190_base_202420_1 vck190 Embedded Accel xilinx.com ...vck190_base_202420_1\xilinx_vck190_base_202420_1.xpfm
xilinx_zcu104_base_202420_1 xd Embedded Accel xilinx.com ...zcu104_base_202420_1\xilinx_zcu104_base_202420_1.xpfm
xilinx_zcu102_base_202420_1 xd Embedded Accel xilinx.com ...zcu102_base_202420_1\xilinx_zcu102_base_202420_1.xpfm
xilinx_zcu102_base_dfx_202420_1 xd Embedded Accel xilinx.com ...ase_dfx_202420_1\xilinx_zcu102_base_dfx_202420_1.xpfm
xilinx_vek280_base_202420_1 xd Embedded Accel xilinx.com ...vek280_base_202420_1\xilinx_vek280_base_202420_1.xpfm
xilinx_vmk180_base_202420_1 xd Embedded Accel xilinx.com ...vmk180_base_202420_1\xilinx_vmk180_base_202420_1.xpfm
xilinx_vck190_base_dfx_202420_1 xd Embedded Accel xilinx.com ...ase_dfx_202420_1\xilinx_vck190_base_dfx_202420_1.xpfm
Cancel
Back
Next


Create Application Component - Linux Hello World
Name and Location > Hardware > Domain > Sysroot > Summary
Select Domain
Choose a domain from the available domains in the selected platform.
Name
Details
linux_ps7_cortexa9
+ create new...
Name linux_ps7_cortexa9
Display Name linux_ps7_cortexa9
OS linux
Processor cortex-a9
Cancel
Back
Next



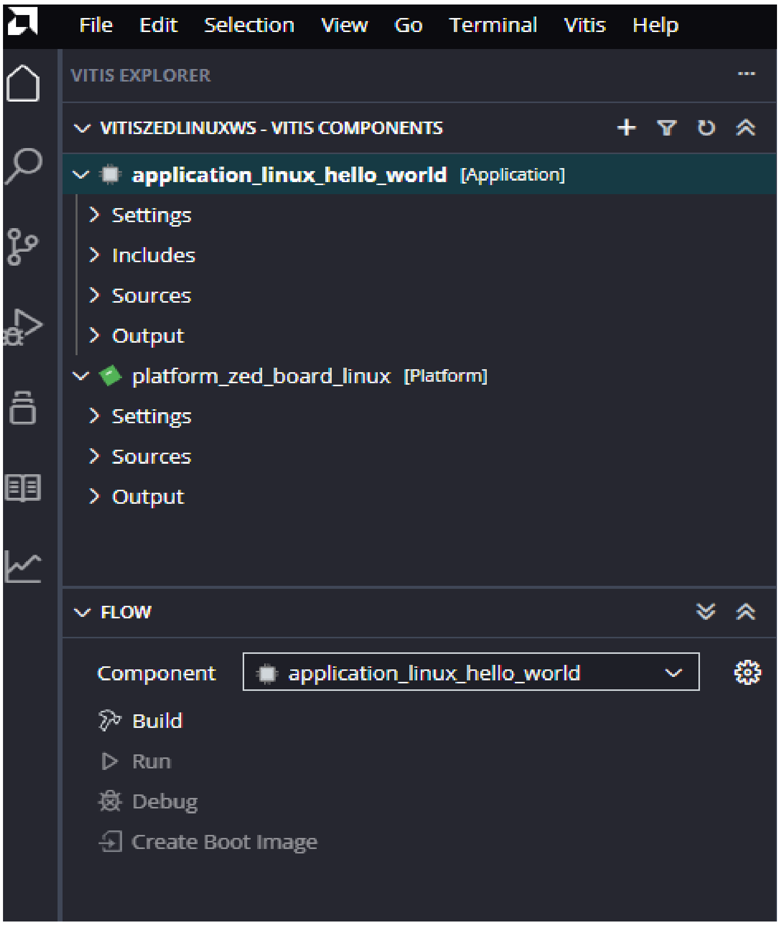


Output Console:

```
19:01:42 INFO  :  Found compiledb at C:\Xilinx\Vitis\2024.2\vitis-server\scripts\compiledb
```

```
19:01:45 INFO  : Generated compile_commands.json at C:\Projects\VitisZedLinuxWs\application_linux_hello_world\_ide
```

4. Building hello world project:



Console Output:

```
--------------------------------------------------------------------------------
[7/16/2025, 7:07:20 PM]: Build for application_linux_hello_world::build with id
'5167207d-f0ec-4dc8-b582-f269e62d0a3a' started.
--------------------------------------------------------------------------------
-- The C compiler identification is GNU 13.3.0
-- The CXX compiler identification is GNU 13.3.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: C:/Xilinx/Vitis/2024.2/gnu/aarch32/nt/gcc-arm-
linux-gnueabi/bin/arm-linux-gnueabihf-gcc.exe - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: C:/Xilinx/Vitis/2024.2/gnu/aarch32/nt/gcc-arm-
linux-gnueabi/bin/arm-linux-gnueabihf-g++.exe - skipped
-- Detecting CXX compile features
```

```
-- Detecting CXX compile features - done
-- Configuring done
-- Generating done
-- Build files have been written to:
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build
 C:/Xilinx/Vitis/2024.2/tps/win64/cmake-3.24.2/bin/cmake.exe
-SC:/Projects/VitisZedLinuxWs/application_linux_hello_world
-BC:/Projects/VitisZedLinuxWs/application_linux_hello_world/build --check-build-
system CMakeFiles/Makefile.cmake 0
 C:/Xilinx/Vitis/2024.2/tps/win64/cmake-3.24.2/bin/cmake.exe -E
cmake_progress_start
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build/CMakeFiles
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build//CMakeFiles/progres
s.marks
 C:/ti/xdctools_3_23_04_60/gmake.exe  -f CMakeFiles/Makefile2 all
 gmake.exe[1]: Entering directory
`C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build'
 C:/ti/xdctools_3_23_04_60/gmake.exe  -f
CMakeFiles/application_linux_hello_world.elf.dir/build.make
CMakeFiles/application_linux_hello_world.elf.dir/depend
 gmake.exe[2]: Entering directory
`C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build'
 C:/Xilinx/Vitis/2024.2/tps/win64/cmake-3.24.2/bin/cmake.exe -E cmake_depends "Unix
Makefiles" C:/Projects/VitisZedLinuxWs/application_linux_hello_world
C:/Projects/VitisZedLinuxWs/application_linux_hello_world
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build/CMakeFiles/applicat
ion_linux_hello_world.elf.dir/DependInfo.cmake --color=
 gmake.exe[2]: Leaving directory
`C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build'
 C:/ti/xdctools_3_23_04_60/gmake.exe  -f
CMakeFiles/application_linux_hello_world.elf.dir/build.make
CMakeFiles/application_linux_hello_world.elf.dir/build
 gmake.exe[2]: Entering directory
`C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build'
 [ 50%] [32mBuilding C object
CMakeFiles/application_linux_hello_world.elf.dir/src/helloworld.c.obj[0m
 C:/Xilinx/Vitis/2024.2/gnu/aarch32/nt/gcc-arm-linux-gnueabi/bin/arm-linux-
gnueabihf-gcc.exe -DVITIS_MAJOR_VERSION="" -DVITIS_MINOR_VERSION=""
-DVITIS_VERSION=""  -g -Wall -Wextra -O0 -g3 -MD -MT
CMakeFiles/application_linux_hello_world.elf.dir/src/helloworld.c.obj -MF
CMakeFiles/application_linux_hello_world.elf.dir/src/helloworld.c.obj.d -o
CMakeFiles/application_linux_hello_world.elf.dir/src/helloworld.c.obj -c
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/src/helloworld.c
 [100%] [32m[1mLinking C executable application_linux_hello_world.elf[0m
 C:/Xilinx/Vitis/2024.2/gnu/aarch32/nt/gcc-arm-linux-gnueabi/bin/arm-linux-
gnueabihf-gcc.exe
"CMakeFiles/application_linux_hello_world.elf.dir/src/helloworld.c.obj"  -o
application_linux_hello_world.elf
 gmake.exe[2]: Leaving directory
```

```
`C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build'
 [100%] Built target application_linux_hello_world.elf
 gmake.exe[1]: Leaving directory
`C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build'
 C:/Xilinx/Vitis/2024.2/tps/win64/cmake-3.24.2/bin/cmake.exe -E
cmake_progress_start
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build/CMakeFiles 0
 Build Finished successfully
 ---------------------------------------------------------------------------------
[7/16/2025, 7:07:24 PM]: Build for application_linux_hello_world::build with id
'5167207d-f0ec-4dc8-b582-f269e62d0a3a' ended.
```

- Looks like we successfully created/built a ZedBoard platform and Linux Hello World application using Vitis!

- Next, we are going to try to launch a debug session from Vitis ....